GigaWhat Energy

GigaWhat Electricity
Controlled procedure

PR-ELEC-020 Permit-to-Work and Sanction-for-Test (Electrical)

Field	Value
Document ID	PR-ELEC-020
Version	7
Effective from	2024-09-02
Review due	2027-09-02
Owner	Electricity Safety Manager
Approved by	Head of Electricity Network Operations
Applies to	All GigaWhat Electricity sites, plant and equipment

FICTIONAL DOCUMENT. Created for the GigaWhat demo. Not for real operational use.

1 Purpose

This procedure sets out how Permits-to-Work and Sanctions-for-Test are prepared, issued, used, transferred, cleared and cancelled on GigaWhat Electricity plant, so that everyone working on equipment knows exactly what has been made safe and where they may work.

2 Scope

This procedure applies to all work and testing on HV equipment at all GigaWhat Electricity sites, and to LV and DC equipment where an Authorised Person (HV) decides a permit is needed to control the risk. Entry to the Kingsmead cable tunnel also needs a confined space entry permit under PR-ELEC-031.

3 References

- PR-ELEC-001 Distribution Safety Rules - Summary for Operational Staff
- PR-ELEC-012 Isolation and Earthing of 11 kV Circuits
- PR-ELEC-015 Switching Programmes on the 11 kV and 33 kV Networks
- PR-ELEC-031 Entry to Kingsmead Cable Tunnel (Confined Space)
- PR-CORP-001 Reporting Incidents and Near Misses
- PR-CORP-004 Lone Working
- HSE HSG250 Guidance on permit-to-work systems
- HSE HSG85 Electricity at work: Safe working practices
- HSE HSR25 Memorandum of guidance on the Electricity at Work Regulations 1989

- Electricity at Work Regulations 1989
- Management of Health and Safety at Work Regulations 1999

4 Definitions

Permit-to-Work (PTW): a safety document stating which equipment has been isolated, proved dead and earthed, and what work may be done at the point of work.

Sanction-for-Test (SFT): a safety document allowing testing that needs earths removed or test voltage applied.

Issuing Authorised Person: the Authorised Person (HV) who issues and cancels the safety document.

Recipient: the Competent Person (HV) who receives the safety document and is in charge of the working party.

Clearance: the recipient's signed statement that work has stopped, everyone has been withdrawn and all Additional Earths have been removed.

Cancellation: the issuing Authorised Person's signed acceptance of the clearance, after which the safety document is no longer in force.

5 Responsibilities

Senior Authorised Person (HV): audits at least five cancelled safety documents each month and approves the cancellation of a lost permit.

Authorised Person (HV): prepares and issues safety documents, shows the recipient the safety precautions on site, and cancels documents.

Competent Person (HV): as recipient, briefs and controls the working party and keeps work within the document's limits.

Control Engineer: issues safety document numbers from the electronic safety document log, records issue and cancellation, and does not instruct any operation that conflicts with a document in force.

Electricity Safety Manager: owns the permit forms and reviews audit findings each quarter.

6 Procedure

6.1 Preparing to issue a Permit-to-Work

6.1.1 Confirm that the equipment has been isolated, proved dead and earthed in accordance with PR-ELEC-012.

6.1.2 Confirm that the keys of all safety locks are in the Safety Key Safe.

6.1.3 Obtain a safety document number from the Control Engineer.

6.1.4 Complete the permit form, stating the equipment by asset ID, the work to be done, and the points of isolation, safety locks and Circuit Main Earths as required by PR-ELEC-012.

6.1.5 State on the permit any adjacent equipment that remains live and how it is marked.

> HOLD POINT: The Authorised Person (HV) must not issue the permit until the Control Engineer has confirmed that the network diagram shows the same isolation and earthing as recorded on the permit.

6.2 Issuing a Permit-to-Work

6.2.1 Meet the recipient at site. Never issue a Permit-to-Work by telephone or radio.

6.2.2 Show the recipient each point of isolation, each Circuit Main Earth and the point of work, and identify adjacent equipment that remains live.

6.2.3 Mark the point of work with barriers and fix Danger notices on adjacent equipment that remains live.

6.2.4 Read the permit to the recipient, who reads it back and confirms they understand its limits.

6.2.5 Have the recipient sign the receipt, hand over the top copy and keep the duplicate.

6.2.6 Tell the Control Engineer the time of issue and the recipient's authorisation number.

> HOLD POINT: The working party must not start work until the recipient has signed the receipt and holds the top copy of the permit at the point of work.

6.3 Working under a Permit-to-Work

6.3.1 As recipient, brief each member of the working party on the permit and have each sign on before work starts.

6.3.2 Keep the permit at the point of work for as long as work continues.

6.3.3 Prove dead and apply Additional Earths at the point of work where needed, in accordance with PR-ELEC-012, and record them on the permit.

6.3.4 Keep the working party within the limits of the permit. If the work or the point of work needs to change, stop and contact the issuing Authorised Person (HV).

6.3.5 Stop work if you, as recipient, have to leave site, unless the permit has been transferred under 6.5.

6.3.6 Sign members of the working party off the permit as they leave site.

6.4 Clearing and cancelling a Permit-to-Work

6.4.1 As recipient, when work is complete or stopped, withdraw the working party, remove tools and materials, and remove all Additional Earths.

6.4.2 Sign the clearance, stating whether the work is complete and whether the equipment is fit for service.

6.4.3 As issuing Authorised Person (HV), inspect the point of work, confirm that all Additional Earths have been removed and sign the cancellation.

> HOLD POINT: No Circuit Main Earth or safety lock may be removed until the permit has been cancelled and the Control Engineer has recorded the cancellation.

6.4.4 Inform the Control Engineer that the permit is cancelled, then remove the Circuit Main Earths in accordance with PR-ELEC-012.

6.5 Transferring a Permit-to-Work

6.5.1 Transfer a permit to a new recipient only when work continues across a shift change or the original recipient must leave site.

6.5.2 Withdraw the working party and have the original recipient sign the transfer section, stating the state of the work and any Additional Earths in place.

6.5.3 As Authorised Person (HV), show the new recipient the safety precautions on site and obtain their signature on the receipt.

6.5.4 Inform the Control Engineer of the transfer.

6.5.5 Do not transfer a permit more than twice. After a second transfer, cancel it and issue a new permit.

6.6 Sanction-for-Test

> WARNING: Testing can involve removing earths and applying test voltages of up to 20 kV to 11 kV cables. Equipment under test is live and can hold a dangerous charge after the test ends.

6.6.1 Issue a Sanction-for-Test only when every Permit-to-Work on the same equipment has been cleared and cancelled. Only one safety document may be in force on the same equipment at a time.

6.6.2 State on the Sanction-for-Test which Circuit Main Earths the recipient may remove for the test.

6.6.3 Mark the test area with barriers and Danger notices, and post a person at any remote end of a cable under test.

> HOLD POINT: The recipient must not remove any earth or apply test voltage until the issuing Authorised Person (HV) has confirmed that every end of the equipment is guarded and nobody is working on it.

6.6.4 After testing, discharge the equipment, reapply every Circuit Main Earth that was removed and clear the Sanction-for-Test.

6.6.5 Cancel the Sanction-for-Test before issuing a new Permit-to-Work on the same equipment.

6.7 Lost permits and emergencies

6.7.1 If a permit is lost or the recipient cannot be found, stop work and inform the Control Engineer.

6.7.2 Cancel a permit without the recipient's clearance only with the approval of the Senior Authorised Person (HV), after physically checking that everyone has been withdrawn and all Additional Earths removed.

6.7.3 If anyone is injured or there is a fire, call 999 and then inform the Control Engineer.

6.7.4 Ask members of the public who report a power cut or damaged power lines to call 105.

6.7.5 Report every incident, near miss or permit irregularity under PR-CORP-001.

7 Records

- Duplicate copies of safety documents, returned to the Control Engineer within seven days of cancellation and kept for six years.
- The electronic safety document log maintained by the Control Engineer.
- Senior Authorised Person (HV) audit records, reviewed quarterly by the Electricity Safety Manager.

8 Revision history

Version	Date	Change
1	2006-04-03	First issue.
2	2009-05-05	Added Sanction-for-Test.
3	2012-09-03	Added transfer of permits at shift change.
4	2015-09-07	Added working party sign-on and sign-off.
5	2018-09-03	Added monthly audit of cancelled safety documents.
6	2021-09-06	Added rule of one safety document in force per item of equipment.
7	2024-09-02	Safety document numbers now issued from the electronic safety document log. Added hold point confirming that the network diagram matches the permit before issue. Limited each permit to two transfers.